



AquaNova

Smart Irrigation System

Arduino-Based Precision Agriculture

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Project Team



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Introduction



What is Smart Irrigation?

A technology-driven system that monitors and controls plant watering automatically using sensors and microcontrollers — no human effort needed.

Why Does It Matter?

Agriculture consumes over 70% of global freshwater. Smart irrigation can cut water usage by up to 40% while improving crop yield.

Our Goal

Design a low-cost Arduino-based system to automate watering, lighting, and monitoring — making agriculture smarter and greener.



Problem Statement



Forgetting to Water

Plants dry out when owners forget regular watering schedules, leading to crop death and wasted planting efforts.



Overwatering

Excess water suffocates roots, causes root rot, and wastes precious water resources unnecessarily.



Insufficient Sunlight

Indoor or shaded plants lack adequate light for photosynthesis, stunting growth and reducing yield.



Water Waste

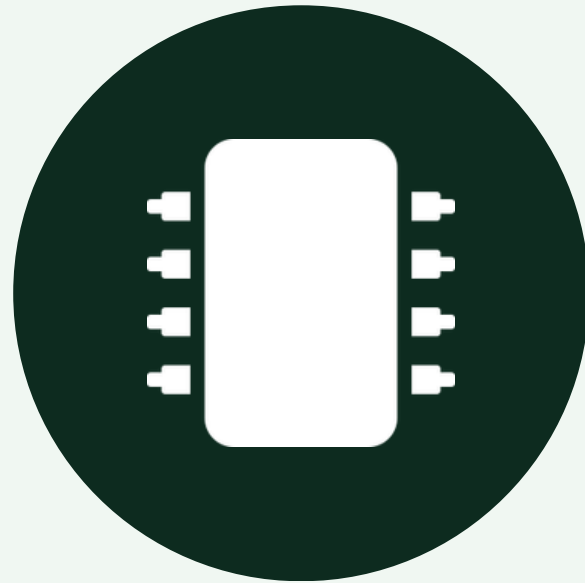
Random, uncontrolled irrigation wastes up to 60% of water with no real agricultural benefit.



Proposed Solution



**Sensors
Read Data**



**Arduino
Processes**



**Pump / Light
Control**



**Optimal
Growth**

Why Our Solution is Better:

- An automated system that monitors soil moisture, temperature, humidity, and light conditions in real-time
- Uses sensors to decide when irrigation is needed
- Operates the water pump only when soil is dry and during daytime
- Controls water flow sensor
- Displays system data and status on an LCD screen
- Saves total water usage using EEPROM



Sensors & Connections (1/3)



Soil Moisture Sensor

ANALOG IN

- Supply: 3.3V – 5V DC
- DRY_THRESHOLD: 700
- WET_THRESHOLD: 500



What It Does:

Measures water content in the soil. When soil is dry (raw > 700), it triggers the water pump. Resets when soil is wet (raw < 500) — Hysteresis logic.

Connection:

→ Arduino Analog Pin A0
Averaged over 10 readings, mapped to 0–100% moisture.



DHT11 Temp & Humidity Sensor

DIGITAL IN

- Temp range: 0–50 °C
- Humidity: 20–90% RH
- Retry: 3 attempts



What It Does:

Reads ambient temperature & humidity. Used to display conditions before/during/after irrigation. Retried 3 times if failed.

Connection:

→ Arduino Digital Pin D7
Read via DHT library with 3-attempt retry logic.



Sensors & Connections (2/3)



LDR Light Sensor

DIGITAL IN

- Pin D4 (INPUT)
- 0 = Bright / Day
- 1 = Dark / Night



What It Does:

Detects daytime vs. nighttime. Irrigation only runs during daytime (light==0). Prevents watering at night to reduce fungal risk.

Connection:

→ Arduino Digital Pin D4
digitalRead: 0 = Bright (day), 1 = Dark (night).



Water Flow Sensor (YF-S201)

INT (D2)

- Target: 0.25 L
- Timeout: 60 sec
- Cooldown: 5 min



What It Does:

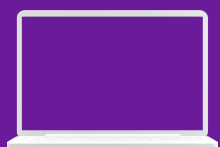
Counts pulses via interrupt to measure water volume. Pump stops when TARGET_VOLUME (0.25 L) is reached or after 60s timeout.

Connection:

→ Arduino Digital Pin D2 (INT0)
attachInterrupt — FALLING edge, 0.25L per 25,000 pulses.



Sensors & Connections (3/3)



LCD

DIGITAL OUT

- I2C Communication (SDA / SCL)
- Address: 0x27
- 16x2 Display

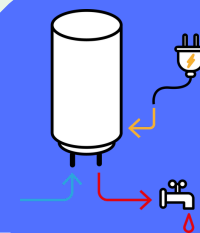


What It Does:

- Displays real-time sensor data
- Shows system status (Pump, Soil, Day/Night)
- Updates only when values change

Connection:

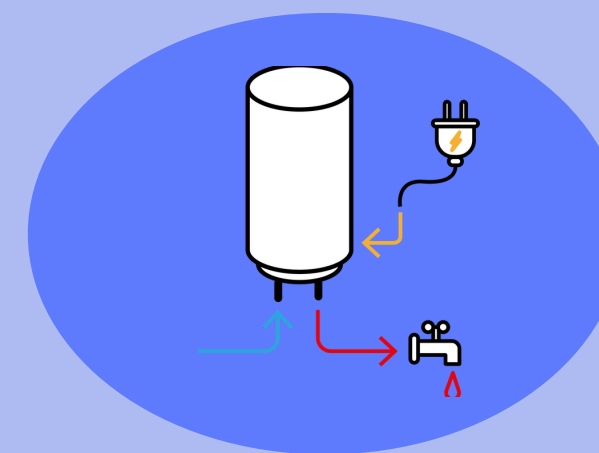
- SDA → A4 (Arduino UNO)
- SCL → A5 (Arduino UNO)
- VCC → 5V
- GND → GND



Water Pump (Relay Controlled)

DIGITAL OUT

- Controlled via Relay Module
- LOW = Pump ON
- HIGH = Pump OFF

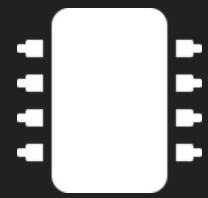


What It Does:

- Irrigates soil when it is dry
- Works only during daytime
- Stops automatically after reaching target water level

Connection:

- Relay IN → D9 (Arduino)
- VCC → 5V
- GND → GND



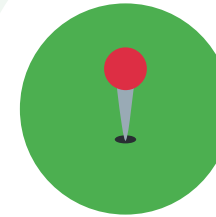
Code Walkthrough (1/3): Pins & Constants

```
// ===== Pin Definitions & Thresholds =====
```

```
#define DHTPIN      7
#define SOIL_PIN    A0
#define LDR_PIN     4
#define RELAY_PIN   9
#define FLOW_PIN    2
```

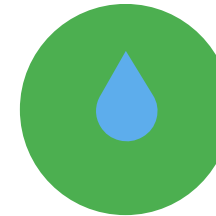
```
const int DRY_THRESHOLD = 700;
const int WET_THRESHOLD = 500;
const float TARGET_VOLUME = 0.25;
const float LITERS_PER_PULSE
    = 0.25 / 25000.0;
```

```
const unsigned long PUMP_TIMEOUT = 60000;
const unsigned long COOLDOWN_MS  = 300000;
const unsigned long SOAKING_MS   = 15000;
```



Pin Mapping

Every sensor/actuator has its own defined pin. RELAY on D9 (active LOW — HIGH = OFF, LOW = ON).



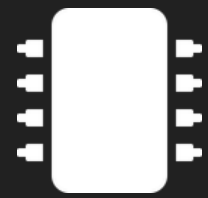
Hysteresis Thresholds

DRY=700 starts irrigation. WET=500 stops it. Gap prevents relay from flickering on/off rapidly.



Timing Constants

PUMP_TIMEOUT=60s safety cutoff. COOLDOWN=5min between cycles. SOAKING=15s water absorption wait.



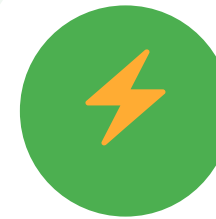
Code Walkthrough (2/3): setup() & Interrupt

```
// ===== Setup & Flow Interrupt =====
```

```
volatile unsigned long pulseCount = 0;
```

```
void flowISR() {  
    pulseCount++;  
}
```

```
void setup() {  
    pinMode(RELAY_PIN, OUTPUT);  
    digitalWrite(RELAY_PIN, HIGH);  
    attachInterrupt(  
        digitalPinToInterrupt(FLOW_PIN),  
        flowISR, FALLING);  
    dht.begin();  
    lcd.init();  
    lcd.backlight();  
}
```



Interrupt Service Routine

flowISR() increments pulseCount on every FALLING edge from the flow sensor pin D2.



RELAY_PIN starts HIGH

HIGH = pump OFF by default. Prevents accidental flooding on startup or reset.



LCD I2C Init

lcd.init() + lcd.backlight() initialises the 16×2 display over I2C address 0x27.

Code Walkthrough (3/3): loop() — Decision Logic

```
// ===== Irrigation Decision =====
```

```
// Hysteresis
```

```
if (soilRaw > DRY_THRESHOLD)
```

```
    soilWasDry = true;
```

```
if (soilRaw < WET_THRESHOLD)
```

```
    soilWasDry = false;
```

```
bool cooldownOK = (millis() -  
    lastIrrigTime >= COOLDOWN_MS);
```

```
bool isDaytime = (light == 0);
```

```
if (soilWasDry &&  
    isDaytime && cooldownOK) {  
    digitalWrite(RELAY_PIN, LOW);  
    // pump until 0.25L or timeout  
}
```



Hysteresis Logic

soilWasDry flag set at raw>700, cleared at raw<500.
Prevents chattering around a single threshold.



Night Guard

isDaytime = (light==0). LDR on D4 blocks irrigation at
night to reduce fungal disease risk.



Cooldown Guard

5-minute cooldown between irrigation cycles prevents
over-saturation and waterlogging.



Complete Wiring Summary

Component	Arduino Pin	Mode	Notes from Code
Soil Moisture Sensor	A0	Analog IN	Avg 10 reads → map to 0-100%
DHT11 Temp/Humidity	D7	Digital IN	3-attempt retry, DHT library
LDR Light Sensor	D4	Digital IN	0=Day (irrigate), 1=Night (skip)
Water Flow Sensor	D2	INT0 FALLING	pulseCount ISR, 0.25L/25000 pulses
Relay Module	D9	Digital OUT	HIGH=OFF (default), LOW=ON (pump)
LCD Display 16×2	SDA/SCL	I2C (0x27)	LiquidCrystal_I2C library
Water Pump	Via Relay D9	Actuator	Stops at 0.25L or 60s timeout



EEPROM: Total water usage saved at address 0 — persists across resets. Validated on boot (0–9999 L range check).



Future Plan

Phase 1



Mobile App Integration

Develop an iOS/Android app to monitor sensor readings in real time and receive alerts when plants need attention.

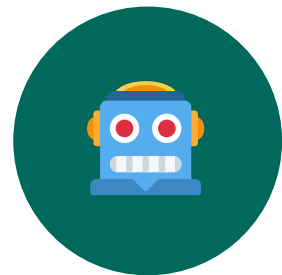
Phase 2



Cloud & AI-Based Prediction

Upload data to cloud and use machine learning to predict irrigation needs based on weather and historical data.

Phase 3



ESP32 & IoT Network

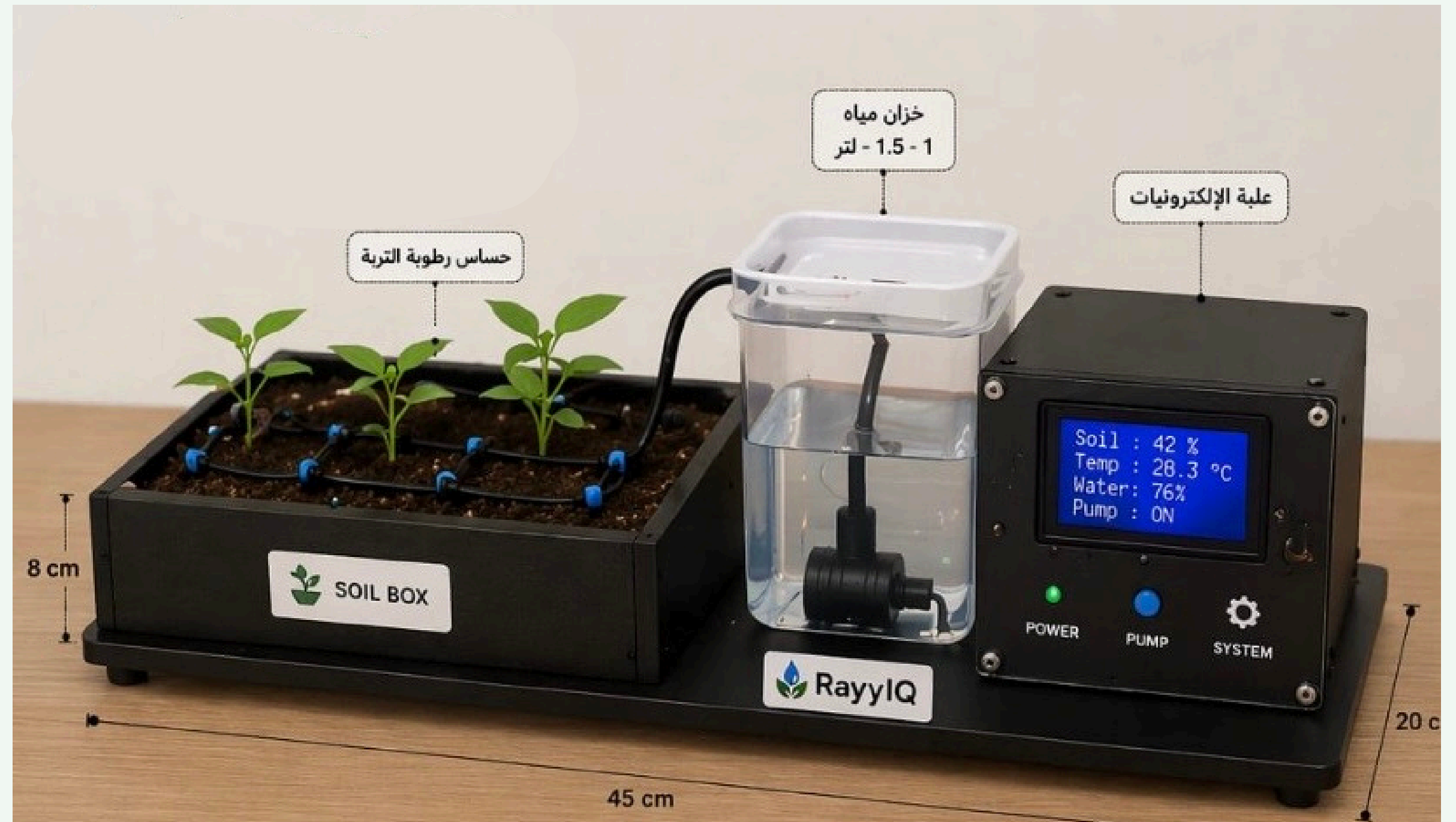
Replace Arduino with ESP32 for Wi-Fi; build a multi-node sensor network covering entire greenhouses remotely.

Phase 4



Solar-Powered Deployment

Integrate solar panels for a fully off-grid system deployable in remote agricultural areas worldwide.





Thank You!

Questions & Discussion

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